

## MARSEILLES, IL, USA

WMO: 744600

Lat: 41.367N

Lon: 88.683W

Elev: 738

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 86-96

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -5.2       | 0.3 | -12.6                         | 2.9 | -3.4 | -6.0 | 4.1 | 1.9  | 28.7                  | 25.4 | 26.8 | 23.9 | 8.6                   | 290  | 0.545 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.5                   | 92.6            | 74.6 | 89.5 | 74.0 | 86.7 | 72.1 | 78.4                | 87.9 | 76.6 | 86.0 | 74.8 | 83.7 | 9.2                  | 230  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 75.8                            | 138.9 | 85.3 | 73.6 | 128.7 | 81.6 | 71.8 | 120.9 | 80.5 | 42.5          | 87.1 | 40.4 | 86.7 | 38.7 | 83.6 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |  |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|--|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |  |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |  |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |  |
| 24.8              | 21.4 | 19.2 | -9.9                       | 95.4  | 6.4                | 3.8 | -14.5                                              | 98.1  | -18.2      | 100.4 | -21.8      | 102.5 | -26.4      | 105.2 |      |  |
|                   |      |      | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |      |  |
|                   |      |      | WB                         | -10.6 | 80.6               | 6.2 | 1.7                                                | -15.1 | 81.8       | -18.7 | 82.8       | -22.2 | 83.7       | -26.7 | 84.9 |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 49.9   | 25.0  | 26.9  | 38.5  | 49.7  | 60.8  | 70.2  | 73.5  | 71.4  | 63.7  | 51.3  | 38.8  | 27.8  |
|                                                                               | DBStd       | 19.78  | 11.76 | 11.49 | 11.09 | 9.77  | 8.90  | 7.20  | 5.89  | 7.01  | 8.07  | 9.02  | 9.74  | 11.07 |
|                                                                               | HDD50       | 3091   | 777   | 648   | 389   | 120   | 10    | 0     | 0     | 0     | 5     | 97    | 358   | 687   |
|                                                                               | HDD65       | 6341   | 1241  | 1068  | 822   | 472   | 193   | 33    | 4     | 20    | 120   | 430   | 786   | 1152  |
|                                                                               | CDD50       | 3065   | 0     | 1     | 34    | 109   | 346   | 606   | 729   | 664   | 415   | 139   | 22    | 0     |
|                                                                               | CDD65       | 841    | 0     | 0     | 2     | 12    | 63    | 190   | 268   | 219   | 81    | 6     | 0     | 0     |
|                                                                               | CDH74       | 8238   | 0     | 0     | 21    | 152   | 664   | 2013  | 2636  | 1989  | 695   | 68    | 0     | 0     |
|                                                                               | CDH80       | 2977   | 0     | 0     | 4     | 40    | 194   | 764   | 1032  | 743   | 192   | 8     | 0     | 0     |
| Wind                                                                          | WSAvg       | 9.3    | 11.0  | 10.5  | 11.3  | 11.3  | 9.7   | 8.0   | 6.6   | 6.3   | 7.2   | 9.1   | 10.8  | 10.1  |
| Precipitation                                                                 | PrecAvg     | 36.3   | 2.2   | 1.5   | 2.3   | 3.2   | 4.3   | 4.2   | 4.3   | 4.2   | 2.8   | 2.9   | 2.8   | 1.8   |
|                                                                               | PrecMax     | 48.5   | 4.5   | 5.3   | 4.5   | 6.3   | 7.3   | 8.5   | 8.1   | 12.4  | 6.3   | 8.1   | 6.1   | 3.0   |
|                                                                               | PrecMin     | 26.5   | 0.4   | 0.6   | 0.8   | 1.5   | 0.6   | 0.8   | 0.3   | 0.6   | 0.7   | 0.4   | 0.3   | 0.3   |
|                                                                               | PrecStd     | 5.6    | 1.0   | 1.1   | 1.2   | 1.3   | 2.0   | 2.2   | 2.0   | 2.7   | 1.7   | 1.9   | 1.6   | 0.7   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 54.3  | 57.7  | 76.2  | 85.5  | 88.5  | 95.0  | 96.4  | 97.9  | 89.6  | 80.9  | 69.5  | 54.4  |
|                                                                               |             | MCWB   | 50.0  | 50.8  | 61.2  | 63.7  | 70.1  | 74.3  | 75.8  | 76.5  | 74.3  | 63.5  | 58.0  | 48.0  |
|                                                                               | 2%          | DB     | 47.1  | 50.9  | 68.9  | 78.6  | 85.4  | 90.9  | 92.6  | 91.8  | 85.3  | 75.2  | 63.3  | 49.6  |
|                                                                               |             | MCWB   | 42.3  | 45.4  | 57.2  | 62.3  | 67.4  | 71.9  | 75.1  | 76.3  | 70.3  | 60.8  | 57.0  | 45.3  |
|                                                                               | 5%          | DB     | 42.2  | 45.3  | 62.6  | 71.9  | 81.5  | 87.8  | 89.3  | 87.5  | 81.7  | 71.1  | 59.1  | 44.9  |
|                                                                               |             | MCWB   | 38.4  | 40.2  | 53.9  | 58.5  | 66.2  | 70.3  | 75.4  | 75.6  | 67.8  | 58.4  | 53.9  | 41.0  |
|                                                                               | 10%         | DB     | 38.4  | 40.8  | 56.2  | 66.3  | 77.3  | 84.6  | 86.0  | 83.7  | 77.9  | 66.8  | 54.2  | 40.1  |
|                                                                               |             | MCWB   | 35.4  | 37.2  | 48.9  | 55.1  | 63.5  | 69.4  | 73.7  | 72.6  | 65.7  | 57.1  | 48.7  | 37.3  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 51.1  | 51.9  | 63.7  | 66.5  | 74.5  | 76.8  | 80.9  | 81.1  | 75.9  | 65.5  | 61.3  | 50.5  |
|                                                                               |             | MCDB   | 54.4  | 56.0  | 73.9  | 80.9  | 85.1  | 88.2  | 91.6  | 89.8  | 86.7  | 74.5  | 63.4  | 52.9  |
|                                                                               | 2%          | WB     | 43.2  | 45.8  | 59.0  | 63.6  | 71.4  | 74.9  | 78.5  | 78.8  | 72.5  | 63.5  | 58.2  | 45.2  |
|                                                                               |             | MCDB   | 46.0  | 51.0  | 65.8  | 76.0  | 80.6  | 87.9  | 88.2  | 87.9  | 82.0  | 73.8  | 63.4  | 49.2  |
|                                                                               | 5%          | WB     | 38.7  | 40.8  | 54.3  | 60.3  | 68.6  | 73.2  | 76.8  | 76.8  | 70.4  | 60.4  | 54.0  | 41.1  |
|                                                                               |             | MCDB   | 41.7  | 44.5  | 63.1  | 70.6  | 76.9  | 84.7  | 86.3  | 85.7  | 78.2  | 68.6  | 59.0  | 44.9  |
|                                                                               | 10%         | WB     | 35.7  | 37.2  | 48.4  | 56.8  | 66.0  | 71.2  | 74.9  | 74.8  | 68.2  | 57.8  | 49.3  | 37.5  |
|                                                                               |             | MCDB   | 38.2  | 40.5  | 56.4  | 64.9  | 74.3  | 81.4  | 83.5  | 82.0  | 75.1  | 65.8  | 54.2  | 39.7  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 13.8  | 14.3  | 17.7  | 21.1  | 21.8  | 21.4  | 19.5  | 19.6  | 20.8  | 20.2  | 15.3  | 13.1  |
|                                                                               |             | MCDBR  | 17.9  | 20.4  | 26.5  | 27.4  | 25.0  | 24.7  | 22.6  | 22.1  | 23.8  | 25.4  | 19.9  | 18.5  |
|                                                                               |             | MCWBR  | 14.8  | 16.1  | 16.9  | 15.9  | 12.8  | 12.0  | 10.4  | 10.2  | 11.9  | 14.6  | 16.3  | 15.0  |
|                                                                               | 5% WB       | MCDBR  | 16.7  | 19.3  | 25.3  | 25.1  | 20.3  | 21.1  | 19.8  | 19.0  | 19.5  | 22.2  | 19.4  | 17.5  |
| MCWBR                                                                         |             | 14.3   | 15.8  | 17.4  | 15.9  | 11.7  | 12.0  | 10.6  | 9.8   | 11.1  | 14.4  | 16.7  | 14.6  |       |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.288  | 0.305 | 0.340 | 0.393 | 0.421 | 0.430 | 0.446 | 0.434 | 0.403 | 0.349 | 0.306 | 0.288 |       |
|                                                                               | taud        | 2.438  | 2.396 | 2.396 | 2.255 | 2.232 | 2.248 | 2.190 | 2.215 | 2.276 | 2.433 | 2.501 | 2.457 |       |
|                                                                               | Ebn at Noon | 276    | 288   | 288   | 278   | 272   | 269   | 263   | 263   | 263   | 268   | 267   | 266   |       |
|                                                                               | Edh at Noon | 24     | 30    | 34    | 42    | 44    | 43    | 45    | 43    | 38    | 29    | 23    | 22    |       |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 533    | 819   | 1145  | 1482  | 1698  | 1929  | 1924  | 1700  | 1433  | 951   | 617   | 453   |       |
|                                                                               | RadStd      | 54     | 103   | 89    | 121   | 134   | 142   | 109   | 99    | 123   | 109   | 54    | 53    |       |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (a)  | (b)                     | (c)   | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) | Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (50 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page